Multimodal Data Management

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Course: Algorithms, Data Structures and Databases

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Project Title: Musical Catalog and Advisor (Educational)

Date: December 2025

1. Introduction and Context

This project was developed as part of the *Algorithms, Data Structures and Databases* course and aims to provide a hands-on introduction to the design and implementation of complete data management systems. Rather than focusing on isolated algorithms or theoretical constructs, the project emphasizes the **end-to-end lifecycle of data**, from raw ingestion to analytical exploitation, following principles commonly used in modern data engineering environments.

The central objective of the project is to demonstrate how core concepts covered in the course—such as structured data handling, data organization, reproducibility, automation, and efficient querying—can be integrated into a coherent and extensible pipeline. To achieve this, the project is framed around a **multimodal music catalog**, a realistic and well-motivated use case in which heterogeneous data sources must be managed jointly.

Music data naturally combines multiple modalities: structured metadata (e.g., track name, artist, album), textual descriptors and labels, and numerical audio-related features derived from signal analysis. Managing such heterogeneous information within a unified system presents challenges that closely mirror those encountered in real-world data platforms. By addressing these challenges, the project serves both as an educational exercise and as a practical illustration of contemporary data engineering practices.

In addition, the project explicitly incorporates **DataOps principles**, including automation, data quality control, traceability, and reproducibility. Through the use of Python-based processing, object storage via MinIO, and vector-based retrieval using ChromaDB, the system provides a realistic yet accessible environment for experimenting with multimodal data management and AI-oriented data exploitation.

2. Project Objectives

The primary goal of this project is to design and implement a **complete multimodal data management pipeline** for a musical catalog and advisory system. This overarching goal is decomposed into the following specific objectives:

Design a multimodal data management system for music data

The system integrates multiple representations of musical content, including structured metadata, textual descriptions, and numerical audio features. By treating these modalities jointly, the pipeline enables richer semantic relationships between tracks than would be possible using a single data type in isolation.

Implement a zonal, DataOps-inspired pipeline architecture

The pipeline is structured into four clearly defined zones—**Landing, Formatted, Trusted, and Exploitation**—each corresponding to a different stage of data maturity. This separation enforces clarity of responsibilities, supports reproducibility, and mirrors industry-standard data lake and data warehouse architectures.

Ensure automation, reproducibility, and traceability
All pipeline stages are executed via automated scripts and configured through environment variables. This design ensures that the entire workflow can be re-run deterministically, facilitating experimentation, debugging, and evaluation.

These objectives directly reflect the theoretical foundations of the course and demonstrate their application to a realistic and non-trivial multimedia data scenario.

3. Data Context and Sources

Domain

The project is situated within the domain of **digital music catalogs and recommendation systems**, an area where data engineering, database design, and machine learning intersect. Modern music platforms rely on heterogeneous data sources to support search, discovery, personalization, and recommendation. These systems must efficiently handle large volumes of structured metadata, user-facing textual information, and low-level audio descriptors.

This domain is inherently **multimodal**. Structured and textual data capture descriptive and contextual aspects of music, while audio-related features encode acoustic and stylistic properties. As a result, music data provides an ideal case study for exploring how multiple data modalities can be managed and related within an unified architecture.

Problem Statement

A common challenge in music data systems is the **fragmentation of information across heterogeneous formats and sources**. Musical metadata, descriptive text, and audio characteristics are often stored and processed independently, limiting the ability to perform integrated analysis or retrieval.

The problem addressed by this project is the absence of a structured, automated, and reproducible pipeline capable of ingesting, processing, validating, and relating multimodal music

data. Without such a pipeline, tasks such as semantic search, similarity analysis, or content-based recommendation become difficult to scale, audit, and maintain.

The project addresses this challenge by designing a **DataOps-oriented pipeline** that unifies multiple music-related modalities and prepares them for downstream analytical and exploratory tasks.

Business Value

Although the project is academic in nature, the concepts and techniques it demonstrates have strong real-world relevance. Efficient multimodal music data management is essential for streaming platforms, digital archives, and music recommendation services.
A well-designed multimodal pipeline can:

- Enable faster and more accurate music discovery.
- Support educational and research use cases in music information retrieval.
- Provide a scalable, auditable foundation for recommendation and advisory systems.

By demonstrating how structured pipelines can harmonize heterogeneous music data, the project highlights the practical value of DataOps methodologies in multimedia applications.

4. Data Management Backbone

Zonal Architecture

The data management backbone of the project is structured according to a **zonal architecture inspired by DataOps principles**. Rather than treating data processing as a monolithic ETL task, the pipeline explicitly separates responsibilities across four zones: **Landing, Formatted, Trusted, and Exploitation**. Each zone represents a different stage of data maturity and enforces a specific set of guarantees regarding data structure, quality, and readiness for analysis.

This design choice is intentional and pedagogical. By separating the pipeline into zones, the system becomes easier to reason about, debug, and extend. Each zone can be executed independently, allowing partial reprocessing without re-running the entire workflow. This mirrors real-world data platforms, where raw data is preserved, transformations are incremental, and analytical layers depend only on validated data.

The zonal architecture also reinforces reproducibility and traceability. At any point, it is possible to inspect the outputs of a given zone and understand how data has evolved from its original raw form to its analytical representation.

Landing Zone

The **Landing Zone** acts as the immutable entry point for all raw data ingested into the system. In this project, the Landing Zone is responsible for:

- Ingesting raw music datasets without altering their original structure.
- Preserving original metadata fields and formats.
- Storing raw outputs in object storage (MinIO) as a persistent source of truth.
- Recording execution metadata through manifest files, including sampling parameters and data availability flags.

No cleaning, normalization, or validation is performed at this stage. This is a deliberate choice: the Landing Zone is designed to maximize **data fidelity**, ensuring that downstream stages can always be reproduced or reconfigured using the same raw inputs. Any issues introduced later in the pipeline can therefore be traced back to this original data snapshot.

From a DataOps perspective, the Landing Zone guarantees **immutability, traceability, and recoverability**, which are essential properties in production-grade data systems.

Formatted Zone

The **Formatted Zone** transforms raw data into a standardized and machine-friendly representation.

At this stage, the pipeline:

- Harmonizes column names and schemas across sources.
- Normalizes textual fields (e.g., track names, artist names).
- Converts audio-related attributes into consistent numeric formats.
- Completes missing audio features using auxiliary datasets when available.

The goal of the Formatted Zone is not to enforce data correctness, but rather **structural consistency**. By the end of this stage, all records conform to a common schema, making them suitable for systematic validation and analysis.

An important design decision in this zone is the separation between **formatting and validation**. The pipeline does not discard data in the Formatted Zone; instead, it prepares the data so that quality rules can be applied explicitly and transparently in the Trusted Zone. This separation avoids premature data loss and improves debuggability.

Trusted Zone

The **Trusted Zone** is where data quality is explicitly enforced. In this zone, the pipeline applies deterministic rules to ensure that the dataset meets minimum standards of reliability before being used for analytical tasks. These rules include:

- Deduplication of records based on normalized track and artist identifiers.
- Filtering of empty or malformed entries.
- Consistency checks on key attributes.

The Trusted Zone produces two critical outputs:

1. A **validated dataset** promoted for analytical use.
2. A **data quality report** summarizing how many records were retained, removed, or modified.

This reporting mechanism is essential for auditability and reproducibility. Rather than silently cleaning data, the pipeline documents its decisions, enabling users to understand and justify the final dataset composition.

From a course perspective, this zone directly reflects concepts related to **data validation, integrity constraints, and controlled data curation**.

Exploitation Zone

The **Exploitation Zone** is responsible for transforming validated data into representations suitable for analytical and retrieval-oriented tasks.

In this project, the Exploitation Zone focuses on:

- Generating vector embeddings from structured and textual data.
- Producing alternative representations of audio features, including semantic textual descriptions derived from numerical attributes.
- Storing embeddings in a vector database (ChromaDB) to enable efficient similarity search.

This zone represents the transition from **data management** to **data exploitation**. The data is no longer being cleaned or validated; instead, it is being used to support higher-level operations such as similarity search, multimodal retrieval, and exploratory analysis.

The separation of this zone ensures that analytical experiments are always performed on a stable and trusted dataset, avoiding contamination from raw or partially processed data.

5. Implementation Summary

Tools and Environment

The project is implemented using an open-source technology stack selected to reflect realistic data engineering workflows while remaining accessible for educational purposes.
Python serves as the core implementation language due to its strong ecosystem for data processing, automation, and rapid prototyping. The use of **Pandas** enables explicit control over data transformations, schema normalization, and validation logic, aligning with course topics related to structured data handling.

MinIO is used as object storage to simulate cloud-based data lakes. Its S3-compatible interface reinforces the abstraction of storage from computation and supports persistent, versionable storage of raw and processed data. This choice allows the pipeline to mimic real-world architectures used in large-scale data platforms.

ChromaDB is employed as a vector database in the Exploitation Zone. This component introduces students to modern retrieval systems based on vector similarity rather than traditional relational queries. The use of ChromaDB enables efficient similarity search and supports experimentation with multimodal embeddings.

Sentence Transformers provide off-the-shelf embedding models for textual data. This choice allows the project to focus on data management and evaluation rather than model training, which is outside the scope of the course.

All components are configured using environment variables, ensuring that the pipeline is decoupled from specific execution environments and remains portable and reproducible.

Pipeline Execution and Orchestration

Pipeline execution is orchestrated through a centralized Python script that sequentially executes each processing zone. This orchestration enforces a strict execution order, ensuring that each stage operates only on data that has reached the appropriate level of maturity.

This design reflects a key DataOps principle: **pipelines should be deterministic, automated, and repeatable**. Manual intervention is avoided, reducing the risk of inconsistent results across runs. If a failure occurs at any stage, execution halts immediately, preventing the propagation of incomplete or invalid data.

The orchestration logic also enables partial re-execution. For example, if formatting rules change, the pipeline can be resumed from the Formatted Zone without re-ingesting raw data. This capability supports iterative development and experimentation.

6. Analytical Tasks

A key objective of the project is not only to manage data, but to **enable meaningful analytical tasks once the data has reached a sufficient level of quality and structure**. The pipeline design ensures that analytical operations are performed exclusively in the Exploitation Zone, using validated and standardized data. This separation between preparation and analysis is fundamental to robust data engineering practices.

The implemented pipeline enables several complementary analytical capabilities, each illustrating different aspects of data representation and retrieval.

Same-Modality Similarity Analysis

The first analytical capability supported by the pipeline is **same-modality similarity analysis based on audio features**. Each musical track is represented as a fixed-length numerical vector composed of low-level audio descriptors such as danceability, energy, valence, acousticness, instrumentalness, loudness, and tempo.

These vectors enable similarity comparisons between tracks using standard distance or similarity measures (e.g., cosine similarity). This type of analysis allows the system to explore whether songs that share similar acoustic properties are perceptually or stylistically related. In particular, the project investigates whether tracks belonging to the same genre tend to be closer in this audio feature space than tracks from different genres.

This task directly reflects core ADSDB concepts related to **vector representations, distance-based retrieval, and similarity search**. It also highlights an important limitation: low-level audio descriptors alone do not fully capture high-level semantic categories such as musical genre. This observation motivates the inclusion of additional modalities and richer representations.

Cross-Modal Retrieval and Multimodal Alignment

Beyond same-modality analysis, the pipeline supports **cross-modal retrieval**, a central theme of multimodal data systems. In this setting, textual metadata (e.g., track information, genre labels, and semantic descriptions) and audio-derived representations are embedded into a shared vector space.

The project explores cross-modal alignment by transforming numerical audio features into natural-language semantic descriptions using a heuristic model. These descriptions are then

embedded using the same text embedding model as the structured metadata. This approach allows textual queries to retrieve audio-based representations, even though no supervised alignment or joint training is performed.

This experiment demonstrates both the potential and the limitations of weakly supervised multimodal alignment. While retrieval performance is limited, the experiment provides valuable insight into how different modalities interact in embedding spaces and underscores the importance of representation choice.

Structured Catalog Exploration

In addition to vector-based retrieval, the pipeline preserves structured metadata throughout all stages. This enables traditional catalog exploration tasks, such as filtering tracks by genre, artist, release year, or other attributes.

The coexistence of structured querying and vector-based similarity search illustrates a key architectural principle: **modern data systems often combine classical database operations with embedding-based retrieval**. The project demonstrates how both paradigms can be supported within a single, well-organized pipeline.

7. Data Quality Management and Validation

Data quality management is treated as a first-class concern in this project and is explicitly centralized in the Trusted Zone. Rather than embedding ad hoc cleaning logic throughout the pipeline, validation is performed in a dedicated stage using deterministic and transparent rules.
The validation process includes:

- Normalization of key textual identifiers to ensure consistent comparison.
- Deduplication of records based on normalized track and artist identifiers.
- Removal of records missing essential attributes required for downstream analysis.

Importantly, validation decisions are **documented rather than hidden**. The Trusted Zone generates a structured data quality report that summarizes how many records were processed, retained, or removed, along with the criteria applied.

This explicit reporting serves multiple purposes. It enables auditability, supports reproducibility, and provides insight into the characteristics and limitations of the final dataset. From an educational perspective, it reinforces the idea that data quality management is an explicit engineering activity rather than an implicit preprocessing step.

8. Operations and Automation

The operational design of the pipeline reflects DataOps best practices, emphasizing automation, configurability, and reproducibility.

All pipeline stages are executed via automated scripts, with configuration parameters provided through environment variables. This approach decouples infrastructure and execution details from application logic, allowing the same codebase to be executed in different environments with minimal modification.

Intermediate outputs are persisted at each zone, both locally and, when possible, in object storage. This persistence enables inspection of intermediate results, supports partial re-execution, and facilitates debugging. Logs and summary reports generated at each stage provide additional visibility into pipeline behavior.

Collectively, these design choices ensure that the pipeline behaves as a **deterministic and repeatable system**, a requirement that is critical in both experimental and production data workflows.

9. Results

The pipeline successfully processes the music dataset through all four zones, producing a validated and structured multimodal dataset ready for analytical use. Several empirical observations emerge from the execution of the pipeline:

- The zonal architecture simplifies debugging and iteration by isolating responsibilities across stages.
- Explicit data validation improves transparency and confidence in the analytical results.
- Same-modality audio similarity analysis reveals significant overlap between genres, indicating that low-level audio features alone are insufficient to capture high-level semantic distinctions.
- Cross-modal retrieval experiments demonstrate the feasibility of aligning textual and audio-derived representations, while also highlighting the limitations of heuristic and weakly supervised approaches.

These observations validate the pipeline design and provide meaningful insight into the challenges of multimodal data representation and retrieval.

10. Conclusions and Next Steps

This project demonstrates how structured pipeline design and DataOps principles can be applied to a realistic multimodal music data management scenario. By explicitly separating ingestion, formatting, validation, and exploitation, the system achieves clarity, reproducibility, and extensibility.

The project highlights the importance of disciplined data management as a foundation for analytical and machine learning tasks. It also illustrates the limitations of relying on a single modality and motivates the use of multimodal representations for richer and more flexible analysis.

In the second part of the project, the focus shifts toward **quantitative evaluation of embeddings and modality alignment**, extending the pipeline from data management into experimental assessment. This progression reflects a natural evolution from data engineering foundations to analytical and modeling challenges, reinforcing the core learning objectives of the course.